Assembler Design options

* One pass Assembler
* Two pass Assembler
* Multipass Assembler.

Two pass Assembler

Assembler converts Assembly language pgm to m/c equivalents

Two pass assembler will require two passes to convert Assembly language pgm to its m/c equivalents

* In Pass 1 it assign address to all symbols
  * after the address is assigned will contain

| Label Name | address |
|---|---|
| RDREC | 1033 |

* When RDREC comes second time then it will say that symbol is already present. & if the symbol is not present then error will occur.

* At end of pass 1 an intermediate file will be occured. which contains symbol and its address

## Pass 2:

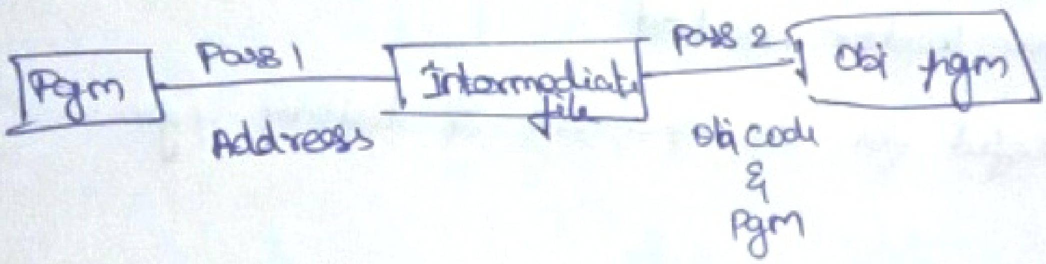
* For In pass 2 object code is written

(Eg) STL RETADR — 14 1033 → m/c equivalent (hexadecimal)

Assembly lang

* The object program will contain
  * Header
  * Text
  * End

  } this object program will loaded & executed.




One pass Assembler.

The main disadvantage of two pass assembler is forward reference

(Eg)
START
STL RETADR
⋮
RETADR Forward reference.

Methods (Types)

* Load & go :
* Proceeds as pass 2

Type 1 (Load & Go)

Eg: STL RETADR

Symbol table

| | |
|---|---|
| RETADR | ~~1033~~ * |
| WRITE | 203B |
| | |

object pgm. H ^ COPY ^ 001000 ^ 00013A

T ^ 14 xxxx ^ 2C 3024 ^ → once value is known * xx value is replaced.

Produce object code directly into the memory.

* NO loader needed
* useful for development & testing pgm.
*

TYPE 2

T ^ 14 0000 ^ 2C3024 - - . → 0000 is ~~replaced~~ replaced as new Text record. once value is known.